

Taxonomical Analysis of 57 multi-domain GPCRs

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13 December 2022

Abstract

Taxonomical/phylogenetic analysis proves to be a useful tool in collecting evidence for or against the existence and functionality of a new protein that has not yet been well-studied. By following some criteria and conditions, close homologues were chosen and their evolutionary relationship was evaluated in the taxonomic context. In the case of multi-domain GPCRs, this was done on the set of mGPCRs that were originally published, a set of 57 proteins that represent clusters that are larger than five similar sequences.

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Results

	Accession	Genus	Close Homologues
1	A0A5J5DEG3	Etheostoma	Dicentrarchus(1), Notolabrus(1), Anarrhichthys(1), Plectropomus(1)
2	A0A7J5XRY7	Dissostichus	Anarrhichthys(1), Plectropomus(1), Etheostoma(1)
3	A0A5N4D1C3	Camelus	Neotoma(1)
4	A0A1A6I049	Neotoma	Galemys(1), Myotis(1), Tupaia(1)
5	A0A1A6GQ15	Neotoma	No mGPCR homologues
6	L5MBD6	Myotis	Lynx(1), Tupaia(1)
7	L9KKD4	Tupaia	Lynx(1), Myotis(1)
8	A0A3L7HS58	Cricetulus	Invalid*
9	L9LEK1	Tupaia	No mGPCR homologues
10	A0A5E4D615	Marmota	No mGPCR homologues
11	A0A3L7HH88	Cricetulus	Invalid*
12	A0A485NDU3	Lynx	No mGPCR homologues
13	A0A091CTM0	Fukomys	Lynx(1)
14	L9KTW2	Tupaia	Lynx(1)
15	A0A3L7HS10	Cricetulus	Invalid*
16	L5K5K3	Pteropus	No mGPCR homologues
17	S9XRQ8	Camelus	No mGPCR homologues
18	A0A3L7GMS1	Cricetulus	Invalid*
19	A0A3L7HEN0	Cricetulus	Invalid*
20	A0A485N3G9	Lynx	Ceratotherium(1), Myotis(1), Nannospalax(1), Ursus(1), Camelus(2)
21	S7N9M8	Myotis	Lynx(1), Camelus(1), Manis(1)
22	A0A3L7HF35	Cricetulus	Invalid*
23	A0A3L7HF00	Cricetulus	Invalid*
24	A0A3L7HFZ0	Cricetulus	Invalid*

25	A0A5E4CY53	Marmota	Sciurus(1)
26	A0A5E4CGQ7	Marmota	No mGPCR homologues
27	S9W9G8	Camelus	Marmota, Galemys, Microtus
28	A0A3L7HXS2	Cricetulus	Invalid*
29	A0A3L7HXQ6	Cricetulus	Invalid*
30	A0A402FIC8	Cricetulus	Invalid*
31	A0A3L7HXX2	Cricetulus	Invalid*
32	B3RF99	Sorex	Galemys(1)
33	A0A7J7ESG3	Diceros	Capra(1)
34	W5NQ75	Ovis	Capra(1), Cervus(1), Sorex(1)
35	A0A3L7GLK1	Cricetulus	Invalid*
36	S7QBY2	Myotis	No mGPCR homologues
37	A0A3L7I3J5	Cricetulus	Invalid*
38	A0A087X729	Poecilia	No mGPCR homologues
39	A0A061I414	Cricetulus	No mGPCR homologues
40	A0A3L7HFH9	Cricetulus	Invalid*
41	A0A061IA73	Cricetulus	No mGPCR homologues

	Accession	Genus	Close Homologues
1	UPI00149064B7	Notolabrus	!8 mGPCRs in Moronidae (bony fish)
2	UPI0010A3F807	Dendiceps	No mGPCR homologues
3	UPI0015E1EB90	Morone	!7 mGPCRs in Moronidae (bony fish)
4	UPI000C6E386B	Xiphophorus	!X. hellerii(1), 6 mGPCRs distributed in Poecillinae
5	UPI0018ACCBAC	Kryptolebias	Poecilia (1)
6	UPI00117FEB6D	Sphaeramia	No mGPCR homologues
7	UPI00165B9D03	Fundulus	No mGPCR homologues
8	UPI0013F2086D	Chelonoidis	No mGPCR homologues
9	UPI00111BD4F7	Nannospalax	Myotis(1), Ursus(1)
10	UPI0003316245	Sorex	No mGPCR homologues**
11	UPI000649B007	Sorex	No mGPCR homologues**
12	UPI000719830D	Ceratotherim	Myotis(2), Ursus(1), Lynx(1)
13	UPI0018901E1C	Talpa	No mGPCR homologues
14	UPI001155D300	Suricata	Lynx(1)
15	UPI000FFD56A3	Vombatus	No mGPCR homologues
16	UPI00148EA50D	Marmota	No mGPCR homologues

Procedure

Each representative sequence of the 57 clusters of multi-GPCRs identified in the original search pipeline were searched against NCBI non-redundant database using protein BLAST. The goal was to identify and analyze the phylogenetic relationship of close homologues to these mGPCRs. The filtering criteria for finding close homologues were as follows:

$$\begin{aligned} SID(x, q) &> 70\% \\ (LEN(q) - 30) &< LEN(x) < (LEN(q) + 150) \end{aligned}$$

where,

x : candidate/matched sequence

q : query sequence (from the 57)

$SID(a, b)$: sequence identity between a and b

$LEN(a)$: length of sequence a

Rationale for the criteria: The lower cutoff of the length was specified in a more strict way than the upper cutoff to ensure that the mGPCR is not cut anywhere to falsely match fragments or single GPCRs. The sequence identity was chosen to be lenient, but nevertheless enable the query to match all of its domains. For example, a sequence identity of 50% of a di-GPCR to a new candidate/homologue, in the best case scenario, constitutes only a match to one of the two GPCR domains in the query protein.

For candidates that passed the above criteria, the genus of the species of the close homologues were noted as well as the number of occurrences in that genus, as seen in the tables above.

Discussion

UniProtKB Proteins

We start our discussion with UniProtKB proteins, which can be identified with the accession IDs that do not contain “UPI” in the beginning of their name. They are relatively better characterized than UniParc (UPI) sequences, but are not as reliable as manually annotated sequences as seen in SwissProt. There is an extent of automatic annotation, as well as some of their structures have been predicted by AlphaFold2. They do get removed or changed from time to time, but this is still relatively infrequent compared to UniParc proteins.

In the whole set of 57 mGPCRs, 41 are from UniProtKB and they can be seen in the first table. 15 out of the 41 (36.6%) are invalid. 26 of the rest of the proteins do not have close homologues in their genus, or their family. Some of these have no matches to predicted proteins and are standalone in the database. For others, they are matches between distant species. These distant species along with the species of the query protein can only be grouped under vertebrates to form a much open group that do not share any obvious evolutionary pressure to have convergent sequences. These open groups are seen again and again throughout the 41 proteins. Rather than matching with a close family member, these proteins are homologous to proteins of very distant species. To take the genus *Lynx* as an example case, potential mGPCRs from bats (*Myotis*), tree shrews (*Tupaia*), rhinoceros (*Ceratotherium*), and meerkats (*Suricata*) are homologous to the ones in *Lynx*. In addition, any of these open groups doesn’t form as many times to support an undiscovered evolutionary pressure to form these multi-domain GPCRs in these unrelated species.

Invalid *Cricetulus griseus* sequences

The “Close homologues” column marked as Invalid* against sequences that match the pattern “A0A3L7_____” are sequences from a WGS sequence of Chinese Hamster that match to the same entry in GenBank (RAZU000000000.2.). When they were discovered and placed in the UniProt database, they probably were splicing alternatives to one DNA sequence, but all the versions were removed to be replaced with the master DNA sequence that is the WGS. As these are no longer reliable as mRNA/protein outputs, they are not considered in the analysis.

UniParc Proteins

The latter table shows the 16 sequences that are under the database UniParc. Most of these sequences have no mGPCR homologues that pass the criteria, and are standalone proteins in the database. 3 sequences that are marked at the beginning with “!” have at least some close homologues in their family.

Xiphophorus

For the genus *Xiphophorus*, the genomic location and the mRNA products were investigated to potentially support the existence of the found mGPCR.

As for the location of the protein, all of them were found to be in the 11th chromosome of the species (3 species in total) in the genus. As the RNA splicing predictions and exon-inferring of non-model organisms lack to a large extent, it is hard to deduce which parts of the DNA sequence is transcribed, spliced out and translated. Going from just the mRNAs (predicted from the genomic sequences), there is not much evidence for or against the mGPCRs as the evidence stands feeble in these under-studied organisms.

Conclusion

In conclusion, the lack of good taxonomic relationships between one of the 57 mGPCRs and identified close homologues support the notion that mGPCRs are artifacts, given that GPCRs could have duplicated and evolved many times after Chordates to form mGPCRs. Even considering the case when GPCRs evolved to form dimers later on in their evolution, meaning forming mGPCRs wasn't possible due to non-functionality, a fundamental molecular signaler like a GPCR receptor should have been dispersed in the family or at least genus of the species, at least for some cases. It could be argued or hypothesized that they still exist, and this remains possible for divergent sequences as conventional repertoire of GPCRs are known to have diverged in sequence to have very low sequence identity in order to accomodate different ligands etc., but we currently lack the methodology to search for multiples of seven transmembrane helices ($n \times 7 \text{ TM}$, $n > 1$).

Alternative splicing

As seen multiple times above in the above tables (e.g. *Cricetulus* sequences, *Sorax* sequences marked with **, and potentially more that were not investigated), one genetic DNA sequence can be computationally parsed to produce more than one mRNAs (which happens in nature as well), but this inference producing more than one GPCR domain is unlikely as most of the high-scoring homologues that have evidence on a transcript-level are always single GPCRs. Moreover, this shortens the list significantly as two different proteins that are representative do not represent two evolutionarily distinct mGPCRs.

To support further the hypothesis of alternative splicing, many other searches in the UniProtKB IDs (data not shown) match 50% to potential mGPCRs in these species, and again not in their proximal family, indicating that a single GPCR domain of the mGPCR is conserved (as seen in high-identity matches),

Unfortunately, given the lack of multi-GPCRs in well-studied organisms, the most likely scenario is that these mGPCR-like sequences are olfactory/chemokine receptors placed back-to-back, and are either spliced out to produce only a single GPCR, or transcribed partially straight away without being spliced out afterwards.

Low quality proteins

As only three of the UniParc, and none of the UniProtKB passed the criteria for close homologues in their close family using the NCBI nr database, it is difficult to assert that these mGPCRs exist.

This list could be re-expanded, and potentially more candidates could be found in the genus or the family of these candidates by looking at their genomes and the genomic sequences that make up these proteins, but that would require arguing against the automatic mRNA deduction of these databases, and I presume this would be hard to do as neither them, or us, know much about the splicing patterns of these organisms.

Evidence that contradicts this possibility starts when single GPCRs with evidence on a transcript level (after splicing, ready to be translated) emerge as close homologues (>90% sequence identity and close family members but are generally 1/2 or 1/3 the length.) supporting that these sequences, if ever analyzed on a transcriptomics level, would probably be just single GPCRs.